

G294-S42-IAP1

HPC/AI Server - Intel® Xeon® 6700/6500 - 2U DP 8 x PCIe Gen5 GPUs Immersion



Key Features

- Immersion Cooling Server
- Supports up to 8 x Dual slot Gen5 GPUs
- Dual Intel® Xeon® 6700/6500-Series Processors
- 8-Channel DDR5 RDIMM / MRDIMM, 24 x DIMMs
- 2 x 10 Gb/s LAN ports via Intel® X710-AT2
- 8 x 2.5" Gen5 NVMe/SATA/SAS-4 bays
- 8 x FHFL PCIe Gen5 x16 for GPUs
- 2 x LP PCIe Gen5 x16
- Dual 3000 W 80 PLUS Titanium redundant power supply

Applications

AI Platform, AI Training Server, AI Inference Server, Visual Computing, High-Performance Computing Server...

Specification

Dimensions	2U (W448 x H87.5 x D850 mm)	Storage	Bottom: 8 x 2.5" Gen5 NVMe/SATA/SAS-4 ^[1]
Motherboard	MS44-G20		^[1] Storage card is required to support SATA and SAS drives.
CPU	Intel® Xeon® 6 Processors - Intel® Xeon® 6700-Series Processors - Intel® Xeon® 6500-Series Processors Dual processor, TDP up to 350 W	PCIe Expansion Slots	8 x FHFL PCIe Gen5 x16, for GPUs 2 x LP PCIe Gen5 x16 *The system is only validated with a uniform GPU model.
OS Compatibility	OS Support List	I/O Ports	Top: 2 x USB 3.2 Gen1, 1 x VGA, 2 x RJ45, 1 x MLAN
Socket	2 x LGA 4710 (Socket E2)	TPM	1 x TPM header (SPI), Optional TPM 2.0 kit: CTM012 1 x PROT connector (only enabled on RoT SKU)
Chipset	System on Chip	Power Supply	Dual 3000 W 80 PLUS Titanium redundant power supply AC Input: 100-240 V *The system power supply requires C19 power cord.
Memory	8-Channel DDR5 RDIMM/MRDIMM, 24 x DIMMs ^[1] [RDIMM] Up to 6400 MT/s (1DPC), 5200 MT/s (2DPC) [MRDIMM] Up to 8000 MT/s ^[2] ^[1] Intel® Xeon® 6700E-Series Processors only support up to 16 x DIMMs. ^[2] MRDIMMs are supported only on select Intel® Xeon® 6 processors with P-cores and only in a 1DPC configuration.	Operating Properties	Operating coolant temperature: 20°C to 35°C Non-operating temperature: -40°C to 60°C Non-operating humidity: 20% to 95% (non-condensing)
LAN	2 x 10 Gb/s LAN (1 x Intel® X710-AT2), Support NCSI 1 x 10/100/1000 Mb/s Management LAN	Packaging Content	1 x G294-S42-IAP1 2 x CPU heatsinks 4 x GPU power cables (16-pin, 250mm) 4 x GPU power cables (16-pin, 280mm) 4 x Carriers
Video	Integrated in ASPEED® AST2600 - 1 x VGA port	Ordering Part Numbers	6NG294S42DR000IAP1*



Learn more at: <https://www.gigabyte.com/Enterprise>

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